

Visual-To-Recipe: Automated Recipe Generation from Food Images

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Motivation & Problem

- **Problem:** Generate detailed, step-by-step cooking recipes directly from images of food.
- **Challenge:** Significant multimodal AI task bridging visual understanding and procedural text generation.
- **Goal:** Develop a deep learning system for this image-to-recipe task.
- **Potential Applications:**
 - Smart kitchen assistants
 - Nutritional analysis tools
 - Culinary education platforms



- Explored multimodal alignment (e.g., CLIP [2]).
- Existing approaches often focus on:
 - Ingredient retrieval
 - Rudimentary instruction generation
- **Our Approach:** Build upon these foundations by incorporating State-of-the-Art (SOTA) open-source Large Language Models (LLMs) fine-tuned on culinary texts to enhance recipe quality and adaptability.

Core Idea: Lightweight Adaptation

- Connect powerful, pre-trained unimodal models (Vision & Language).
- Keep large models **frozen** to leverage their extensive knowledge.
- Train only a minimal **adapter** module between them.
- **Advantage:** Significantly less computation and data needed compared to full fine-tuning.

Main Approach: Prefix Tuning Adaptation

1. Frozen Image Encoder:

- im2recipe model [3]
- ResNet-50 backbone
- Extracts visual embedding
 $\mathbf{z} \in \mathbb{R}^{1024}$
- *Parameters frozen*

2. Frozen Language Model:

- Mistral-7B-Instruct-v0.2 [5]
- Quantized (4-bit via BitsAndBytes [6])
- LLM hidden dim $d_{LLM} = 4096$
- *Parameters frozen*

3. Trainable Prefix Adapter (f_θ):

- Inspired by Prefix Tuning [7].
- Maps image embedding $\mathbf{z}$ to a sequence of k prefix vectors:

$$\mathbf{P} = [\mathbf{p}_1, \dots, \mathbf{p}_k], \quad \mathbf{p}_i \in \mathbb{R}^{d_{LLM}}$$

- Structure: Normalize $\rightarrow$ Project $\rightarrow$ Transformer Encoder $\rightarrow$ Pool $\rightarrow$ Project $\rightarrow$ Reshape.
- Key Hyperparameters: k (e.g., 4), hidden dim (e.g., 512), adapter layers (e.g., 2).
- *Only adapter parameters θ are trained.*

4. Visual Conditioning:

- Prepend prefix vectors to prompt embeddings $\mathbf{E}_{prompt}$:

$$\mathbf{E}'_{prompt} = [\mathbf{p}_1, \dots, \mathbf{p}_k, \mathbf{e}_1, \dots, \mathbf{e}_N]$$

- The prefix acts as a learned "visual prompt" guiding the frozen LLM.

5. Training Objective:

- Standard autoregressive cross-entropy loss.
- Train adapter θ to produce prefixes $\mathbf{P}$ that help the frozen LLM predict the ground-truth recipe text $(x_{N+1}, \dots, x_T)$.

$$\mathcal{L}_{\text{Adapter}} = - \sum_{t=N}^{T-1} \log P_{\phi}(x_{t+1} \mid x_{1:k+N+t}, \mathbf{P}, \mathbf{E}_{\text{prompt}})$$

- ϕ denotes the frozen LLM parameters.

6. Inference:

- 1 Image $\rightarrow$ Frozen Encoder $\rightarrow \mathbf{z}$
- 2 $\mathbf{z} \rightarrow$ Trained Adapter $f_{\theta} \rightarrow \mathbf{P}$
- 3 Text Prompt $\rightarrow$ Embeddings $\mathbf{E}_{\text{prompt}}$
- 4 Form $\mathbf{E}'_{\text{prompt}} = [\mathbf{P}, \mathbf{E}_{\text{prompt}}]$
- 5 $\mathbf{E}'_{\text{prompt}} \rightarrow$ Frozen LLM $\rightarrow$ Generate Recipe Autoregressively

Alternative Lightweight Approach

- Focus: Further computational efficiency during training.
- Components:
 - Frozen Vision Encoder: EfficientNet-B0 [8] (1280-dim embedding)
 - Frozen LLM: Quantized DeepSeek-R1-Distill-Llama-8B [9] (4096-dim hidden)
- Mechanism:
 - Trainable lightweight projection layer (adapter).
 - Maps 1280-dim image embedding to 4096-dim vector.
 - **Adds** this vector to the embedding of the *initial* prompt token (e.g., "Recipe:\n").
- Training: Optimize only the projection layer parameters via cross-entropy loss.
- Inference: Image $\rightarrow$ Encoder $\rightarrow$ Adapter $\rightarrow$ Add to Prompt Token $\rightarrow$ LLM Generate.

Alternative Approach Sample

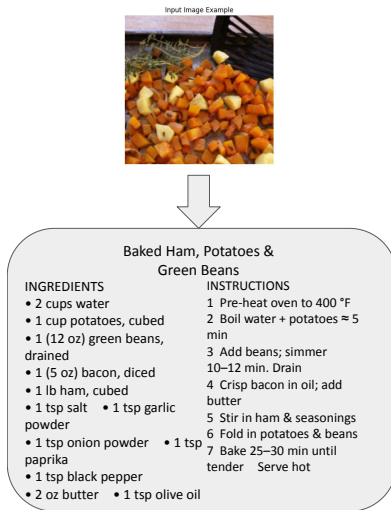


Figure: Sample result from the alternative lightweight approach (shows promise but inaccuracies).

Experimental Settings (Prefix Adapter Method)

- **Dataset:** Recipe1M [3, 1]
 - Used pre-processed subset (*'combined_training_data.pkl'*) containing image tensors and formatted recipe text.
 - Standard Train/Val/Test splits.
- **Models Recap:**
 - Image Encoder: Frozen 'im2recipe' (ResNet-50, 1024-dim out).
 - LLM: Frozen 'Mistral-7B-Instruct-v0.2' (4-bit NF4, 4096-dim hidden).
 - Adapter: Trainable 'PrefixAdapter' ($k = 4$ prefixes, 512 hidden dim, 2 layers, 0.1 dropout).
- **Training:**
 - Optimizer: AdamW ($lr = 1 \times 10^{-5}$).
 - Epochs: 3.
 - Batch Size: 4.
 - Max Sequence Length: 384 tokens.
 - Mixed Precision Training ('torch.cuda.amp').
- **Hardware:** Single NVIDIA A6000 GPU (48GB VRAM).
- **Evaluation Metrics:** BLEU, ROUGE-1, ROUGE-2, ROUGE-L.

Quantitative Results (Prefix Adapter Method)

Performance on Recipe1M Test Set Subset:

Table: Automatic evaluation metrics.

Method	BLEU	ROUGE-1	ROUGE-2	ROUGE-L
Ours	4.35	26.33	4.04	14.57

- Modest scores, but indicate signal. ROUGE-1 and ROUGE-L show reasonable overlap.

Training Loss:



Qualitative Results (Prefix Adapter Method)



Chocolate Chip Walnut Cookies

Ingredients:

- * 1/2 cup butter
- * 1/2 cup sugar
- * 1 egg
- * 1 teaspoon vanilla extract
- * 1 1/2 cups all-purpose flour
- * 1/2 teaspoon baking soda
- * 1/2 teaspoon salt
- * 1 cup semisweet chocolate chips
- * 1 cup chopped walnuts

Instructions:

1. Preheat oven to 350 degrees F (175 degrees C).
2. In a large bowl, cream together the butter and sugar until smooth.
3. Beat in the egg and vanilla extract.
4. In a separate bowl, stir together the flour, baking soda, and salt.
5. Gradually blend into the butter mixture.
6. Mix in the chocolate chips and walnuts.
7. Drop by rounded spoonfuls onto ungreased cookie sheets.
8. Bake for 8 to 10 minutes in the preheated oven, or until edges are nicely browned.
9. Cool on wire racks.

- **Feasibility Demonstrated:** Lightweight adaptation connecting frozen models can generate structured recipe text from images.
- **Learning Confirmation:** Decreasing loss and quantitative/qualitative results show the adapter learned to map visual features meaningfully for the LLM.
- **Generation Quality:**
 - Core structure (ingredients, steps) often present.
 - Struggles with fine-grained details and ingredient accuracy.
 - Visual conditioning via prefix might need more capacity or better integration.
- **Contribution:** Showcased a computationally efficient alternative to full fine-tuning for multimodal generation.
- **Trade-off:** Efficiency comes at the cost of potentially lower generation quality compared to end-to-end trained models. The alternative method (Fig 2) highlights this further.

Conclusion & Future Work

Conclusion:

- Addressed image-to-recipe generation using frozen vision/language models + lightweight trainable adapter.
- Demonstrated feasibility: System generates structurally coherent recipes conditioned on visual input.
- Minimal parameter tuning required, making it computationally efficient.

Limitations & Future Work:

- Improve accuracy of generated details (ingredients, quantities, steps).
- Reduce hallucinations.
- Explore stronger vision-language alignment techniques within the lightweight framework.
- Investigate explicit ingredient recognition modules.
- Refine adapter architecture (more layers, different pooling, etc.).

Paves the way for efficient multimodal generation in domains like culinary

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